

IBRAHIM ROGERS

MECHANICAL ENGINEERING | TEST SYSTEMS & METROLOGY | HARDWARE DEVELOPMENT

B.S. Mechanical Engineering, Northeastern University · Expected May 2028
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SELECTED WORK

01	Liquid Valve Endurance Tester & Leak-Test Fixture	Parker Hannifin · 2026
02	Contamination Root-Cause Investigations	Parker Hannifin · 2026
03	Packaging Redesign & Machine-Vision Inspection	Parker Hannifin · 2026
04	Refreshable Braille Tactile Display	Forge · 2026
05	Vision-Guided Pan/Tilt Tracking System	Independent · 2025-26
06	Gamma Shielding Test System	Melagen Labs · 2024
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Liquid Valve Endurance Tester & Leak-Test Fixture

Quality & Sustaining Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

OVERVIEW

Context: Precision fluid-control valves for medical and life-science systems. My work spans test-system design, fixture design, and root-cause investigation on returned product.

My Role: Requirements definition and hydraulic architecture for a closed-loop endurance tester; mechanical design of a supplier-side leak-test fixture.

CLOSED-LOOP DI-WATER VALVE ENDURANCE TESTER

Converted conflicting stakeholder inputs from quality, manufacturing, and engineering into a single defined test-system scope, then developed the architecture and produced build-ready drawings, schematics, and P&IDs to match.

Test capacity	30–60 valves cycled simultaneously
Flow per valve / aggregate	0.5 L/min / 15–30 L/min
Operating pressure	120 psig
Fluid temperature control	±5°F of ambient
Working fluid	DI water (continuous recirculation)

Sized pump, chiller, reservoir, manifold, and instrumentation, and integrated pressure control, relief protection, filtration, and a chiller bypass loop for thermal management under continuous recirculation. Architecture reserves interfaces for future multi-fluid capability without redesigning the downstream test rack.

PRESSURE-DECAY LEAK-TEST FIXTURE

Designed a fixture in Autodesk Inventor for supplier-side incoming inspection of a molded fluid-control component.

Leak sensitivity requirement	< 0.1 sccm at the orifice
Trapped volume	< 0.5 cc
Method selected	Pressure decay (over make-up flow)
Seal architecture	Guarded three-port
Probe sizing basis	Worst-case tolerance (±0.001 in) + MMC/LMC

DESIGN REASONING

Pressure decay was selected over steady-state make-up flow because commercially available flow instrumentation could not resolve the sub-0.1 sccm requirement at acceptable cost. Locating and sealing functions were separated so the probe seals against the orifice without over-constraining a molded part with real dimensional variation.

Contamination Root-Cause Investigations

Quality & Sustaining Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

OVERVIEW

Led two independent root-cause investigations into recurring return failures on precision fluid-control valves, each with a distinct contamination mechanism and a separate population of returned parts.

INVESTIGATION A — ORIFICE CORROSION

Scope	500+ returned valve bodies
Incidence	Affecting 70%+ of returns
Variables isolated	Supplier, stock, cleaning, floor process
Status	Mitigation validated; ongoing failure-analysis report tracking findings

Designed isolation experiments across each candidate variable rather than inspecting finished returns alone, so the investigation could distinguish a supplier-side cause from a downstream cleaning or floor-process cause instead of guessing from symptom alone.

INVESTIGATION B — FOREIGN-OBJECT CONTAMINATION

Scope	100+ returned units
Method	Teardown and comparative inspection against conforming parts

A separate contamination mechanism from Investigation A, requiring its own teardown protocol and comparison baseline rather than reusing the corrosion investigation's isolation variables.

WHAT I TOOK FROM IT

- **Not every contamination case has the same root cause.** Treating these as two distinct investigations, rather than one aggregate "contamination" bucket, was what let each one get isolated correctly.
- **Comparative inspection needs a real baseline.** Conforming parts had to be inspected alongside returns, or "different" has nothing to be different from.
- **A report that stays open is more useful than one that closes early.** The corrosion investigation's findings are tracked in an ongoing report specifically because mitigation validation is a process, not a single conclusion.

Packaging Redesign & Machine-Vision Inspection

Quality & Sustaining Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

PACKAGING REDESIGN — TRANSIT DAMAGE ON A LOW-VOLUME COMPONENT

Trigger	Customer-reported transit damage
Part profile	Low-volume, high-value (fewer than 500 units/year)
Cost constraint	Fix bounded to $\leq 10\%$ of unit margin

The binding constraint on this project was economic, not technical: a packaging fix that eliminated damage but cost more than the margin it was protecting would not have shipped. Design work had to converge on a solution effective enough to stop the transit failure mode while staying inside the defined cost ceiling — a different kind of constraint than the tolerance- and flow-driven design work elsewhere in this portfolio, but no less real.

MACHINE-VISION INSPECTION SYSTEM — IN DEVELOPMENT

Goal	Flag defects on a production cell before shipment
Stack	Python, OpenCV
Status	Currently in development

This project extends the computer-vision work from my independent pan/tilt tracking system (see page 05) directly into a production application — the underlying detection and image-processing skills transfer, but a production inspection system has to run unattended against real part variation rather than tracking a fiducial marker under controlled conditions, which is the harder problem this project is currently solving.

WHY THESE TWO BELONG TOGETHER

Neither project is a pure mechanical-design exercise. The packaging fix is a cost-constrained design problem; the vision system is a software problem applied to a manufacturing floor. Both reflect the same pattern as the rest of my work at Parker — the actual constraint is rarely only the physics.

Refreshable Braille Tactile Display

Hardware Product Lab — Forge, Northeastern University · Jan 2026 – Jun 2026

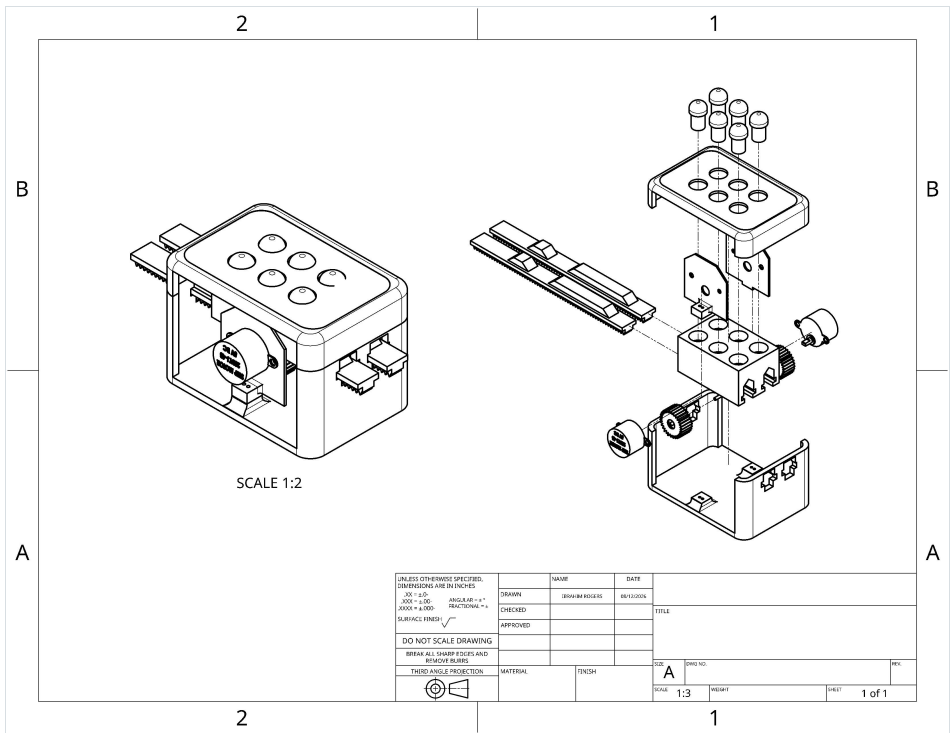
OVERVIEW

Challenge: Design a low-cost, mechanically actuated Braille cell that survives thousands of actuation cycles while holding pin position accurately enough to stay tactilely readable.

My Role: Mechanical design of the actuation mechanism and enclosure, full drawing package, clearance and tolerance-stack iteration, prototype fabrication, and cycle testing.

Approach: A motor-driven rack advances a stepped cam profile that lifts six dome-topped pins through an inner letter-housing insert; the outer enclosure carries the motor, gear, rack, and guide structure around that insert. Modeled in Onshape and released as a full imperial drawing package, with clearances refined iteratively against physical failure modes rather than analysis alone.

FULL ASSEMBLY — EXPLODED & ISOMETRIC

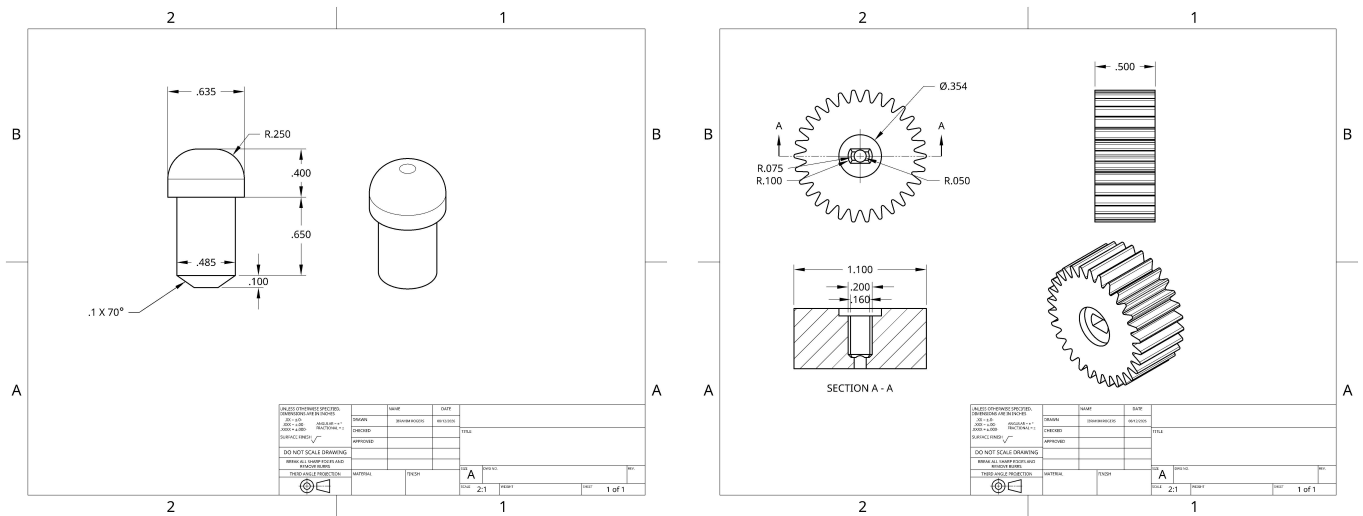


Top-level assembly drawing. Left: assembled enclosure (Scale 1:2). Right: exploded view (Scale 1:3) showing the six dome pins, letter-housing insert, guide bracket, pin-guide block, drive motor and pinion, rack, and enclosure top/bottom. Self-authored release drawing, dimensioned in inches per standard third-angle projection.

OPEN ITEM

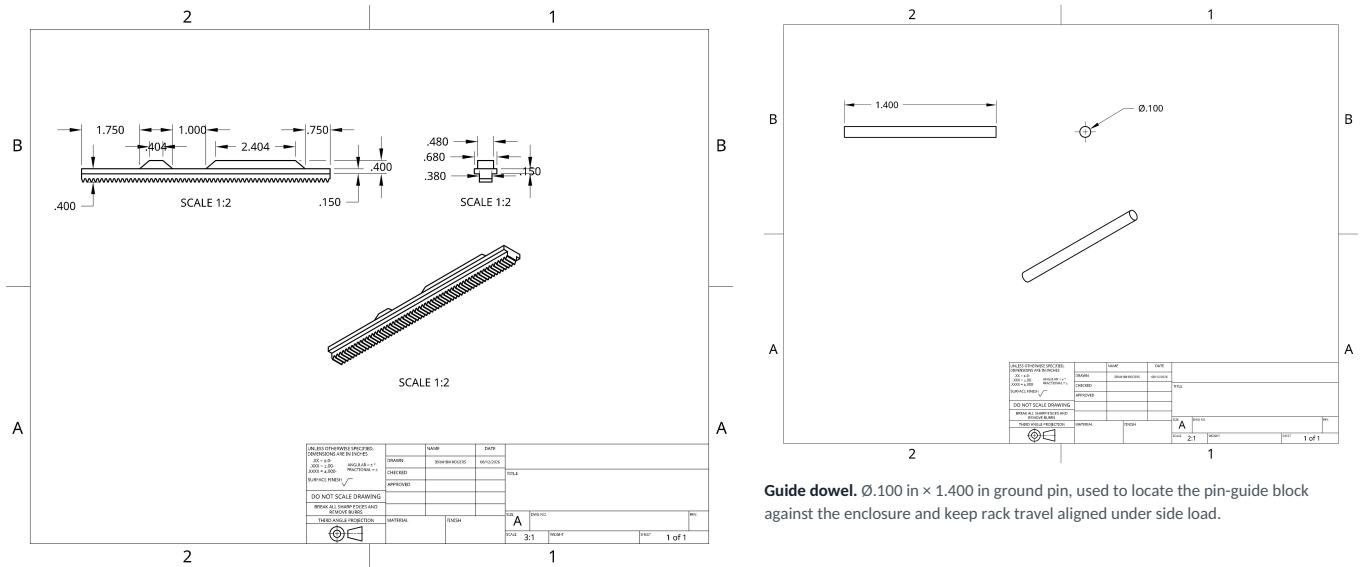
This enclosure (roughly $5.0 \times 3.465 \times 2.5$ in) houses the compact letter-housing insert shown on the following pages; it is not the same part as the insert itself. The envelope figure elsewhere in this portfolio describes the insert, not this outer enclosure — worth double-checking that distinction is stated consistently everywhere this project is referenced.

ACTUATION TRAIN — PIN, GEAR, RACK, DOWEL



Dome pin. Ø.485 in shaft with a Ø.635 in domed head (R.250 radius) and a .1 x 70° lead-in chamfer at the tip. Shaft rides in the Ø.500 in bore of the pin-guide block on the next page — a .015 in diametral clearance for the actuation stroke.

Drive pinion. Ø1.100 in spur gear with a Ø.354 in bore and a keyed drive slot (.200/.160 in) for the motor shaft. Root/fillet radii (R.050–R.100) sized for the FDM-printed tooth profile rather than a cut-gear assumption.



Stepped rack. Linear rack with two lead-in cam profiles at a 1.000 in pitch (Scale 1:2), plus a keyed drive tab (.480/.680/.380 in, Scale 1:2) that engages the pin it lifts. This is the toleranced release version of the stepped-rack concept modeled earlier in Onshape.

Guide dowel. Ø.100 in x 1.400 in ground pin, used to locate the pin-guide block against the enclosure and keep rack travel aligned under side load.

Technical drawing of a 6-hole metal plate. The drawing includes a top view, a side view, and an isometric view. The top view shows a rectangular plate with a width of 5.000 and a height of 3.465. There are six circular holes arranged in a 2x3 grid. The distance between the centers of the holes is 0.650. The side view shows a thickness of 0.197. The isometric view shows the plate with a depth of 4.094 and a height of 0.929. A detail view of the corner shows a radius of R.197.

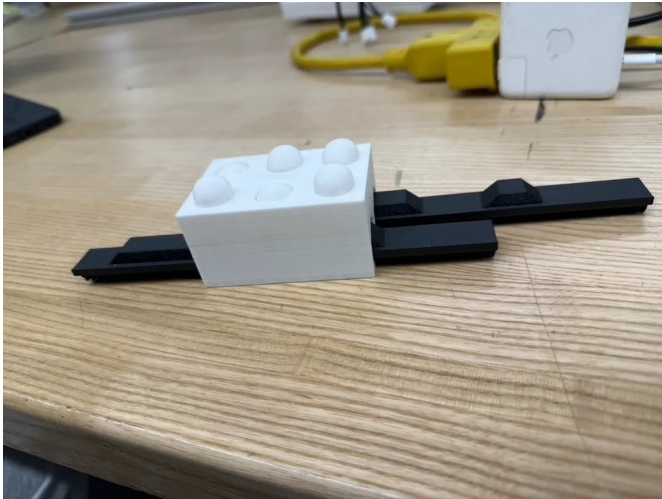
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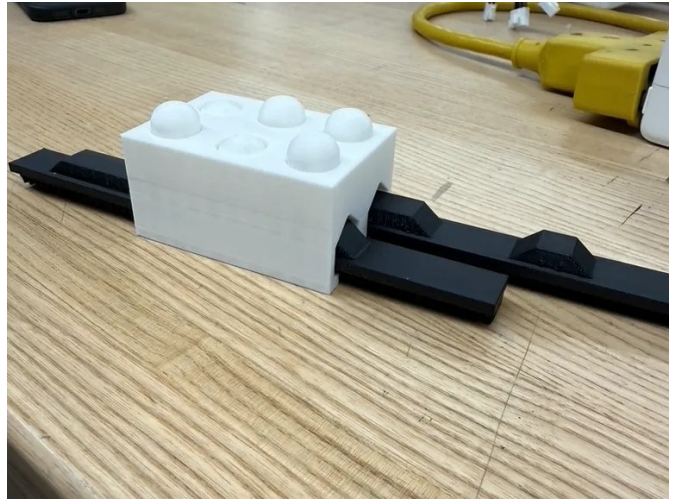
KEY PROBLEM — TOLERANCE STACK-UP UNDER CYCLIC LOADING

- **Clearance is not just a manufacturing number.** It has to absorb print tolerance *and* accumulated wear, not just nominal fit.
- **Material choice is a lifecycle decision.** PLA was fine for form checks and wrong for cyclic loading; that only showed up in testing.
- **Track failures, don't guess.** Logging jam events by condition pointed at side-load binding specifically, which is what made the fix targeted instead of speculative.
- **Design for assembly.** Consolidating cam and guide features cut part count from 12 to 8 and simplified the assembly sequence.
- **Cross-domain constraint.** Worked with the electrical side to keep actuation force inside servo torque limits while still meeting tactile force requirements.

PHYSICAL PROTOTYPE



Assembled prototype — side view. White housing carries the pin actuation mechanism and rides on the black guide rails, which provide lateral constraint. This view shows rail engagement and the travel available to the rack.



Same prototype — front three-quarter view. Shows the six-dot face and the housing-to-rail interface where side-load binding originated. Lead-in chamfers on the rails were added at this interface to ease pin entry.

This early PLA/PETG prototype predates the released drawing package on the preceding pages; it is the build that surfaced the side-load binding described below.

Vision-Guided Pan/Tilt Tracking System

Independent Project · Dec 2025 – Present

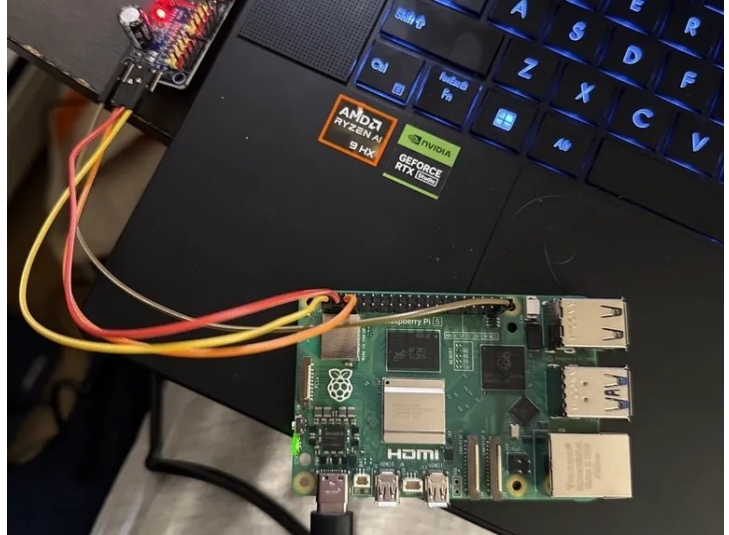
OVERVIEW

Goal: Build a closed-loop target tracking platform — mechanical gimbal, electronics, and control software — that holds a marker centered in frame and recovers on its own when the target is lost.

System: Raspberry Pi 5, camera module, PCA9685 servo driver, dual-servo pan/tilt gimbal, OpenCV ArUco detection, Python control logic, Flask live video stream.



Pan/tilt gimbal. Custom two-axis 3D-printed housing with direct-drive servo coupling — no gear train, which removes backlash from the positioning path.



Electronics integration. Raspberry Pi 5 driving the PCA9685 servo controller over I²C, with servo power kept on a separate rail from compute power to keep actuation transients off the Pi.

MECHANICAL DESIGN DECISIONS

- **Backlash:** direct-drive servo coupling with a press-fit interface, eliminating gear train slop from the control loop.
- **Center of mass:** camera placement chosen to reduce servo torque load and improve response during fast target movement.
- **Rigidity:** gimbal frame reinforced to limit flex under rapid acceleration so the camera stays aligned during dynamic tracking.
- **Cable routing:** integrated to avoid wire interference across the full pan/tilt range.

INTEGRATION DEBUG — CAMERA NOT ENUMERATING

PROBLEM

No frame acquisition and no video feed. From the outside the whole system looked dead, with no obvious indication whether the fault was mechanical, electrical, or software.

PROCESS

- Used `raspistill` to separate camera hardware faults from software faults.
- Verified I²C device enumeration to confirm the Pi was seeing devices on the bus.
- Found improper CSI ribbon seating compounded by outdated camera firmware — two independent faults presenting as one symptom.
- Updated firmware and added error handling in Python for graceful frame-drop recovery.

LESSON

Integration bugs concentrate at interface boundaries. Isolating with diagnostic tools beat guessing at the failure location, especially with two faults stacked.

CONTROL IMPLEMENTATION

- PID loop on pan and tilt axes to smooth servo motion and suppress oscillation around the target; gains tuned empirically on hardware.
- Automated re-acquisition via spiral search when the marker leaves frame or is occluded.
- Error handling for dropped frames, I²C communication failures, and servo limit violations.

WHAT I TOOK FROM IT

- **Stage the integration.** Camera alone, then servos alone, then detection, then full loop. Bringing it up in stages saved hours over debugging everything at once.
- **Textbook PID isn't enough.** Real servo dynamics, friction, and loop latency meant the gains had to come from testing, not from the equations.
- **Cross-domain debugging is the actual skill.** Mechanical slop, power distribution, and software timing all produced similar-looking tracking errors.

Gamma Shielding Test System

Co-Founder & COO — Melagen Labs, VC-backed aerospace materials startup · Jan 2024 – Sep 2024

OVERVIEW

Challenge: Validate the shielding effectiveness of a melanin-HDPE composite against an aluminum reference using gamma radiation, with limited lab equipment and a startup budget.

My Role: Co-designed the test fixture, developed the experimental protocol for repeatable measurement, ran the testing, and performed the statistical analysis.

Key Result: Attenuation comparable to aluminum at roughly one-third the density, supported by ANOVA.

TEST SYSTEM & SETUP



Complete test system. Fixed-geometry fixture holds a repeatable Cs-137 source-to-detector distance. A modular sample holder allows quick swaps between composite thicknesses without disturbing the measurement geometry.

DESIGN REQUIREMENTS

- **Repeatability:** fixed source-to-detector positioning to minimize geometric variation between runs — the dominant error source in this measurement.
- **Modularity:** quick-change sample mounting to test multiple thicknesses and compositions efficiently.
- **Safety:** Cs-137 source containment and shielding per lab safety protocol.
- **Low-cost fabrication:** built from lab-available materials within startup budget.

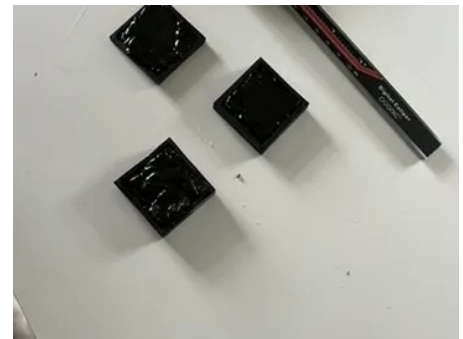
SAMPLE PREPARATION & MEASUREMENT



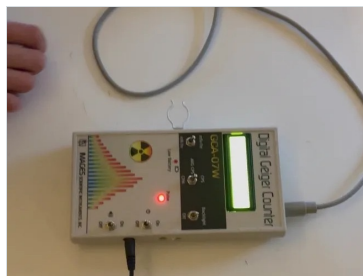
Material preparation. Melanin powder measured to a controlled mass so concentration stayed consistent batch to batch.



Sample fabrication. Composite cast in aluminum pans for consistent thickness, under controlled material-handling protocol.



Test specimens. Composite plates at varying thickness, measured and tested against aluminum references.



Radiation measurement. Calibrated Geiger counter at fixed distance from the shielded source, logging count rate for each shielding condition against an aluminum baseline.

ENGINEERING PROCESS

- **Fixture design for repeatability:** hand-positioning produced unacceptable run-to-run measurement scatter. Adding fixed mechanical alignment features cut that variation substantially and was the single biggest improvement to data quality.
- **Rapid iteration under constraints:** when the first fixture drifted after repeated sample changes, redesigned the mounting within days using available shop tools rather than waiting on ideal equipment.
- **Statistical validation:** ANOVA across material types and thicknesses, with regression modeling of shielding effectiveness against melanin concentration.
- **Uncertainty analysis:** documented contributions from source activity calibration, detector efficiency, and geometric positioning so the results were defensible to outside readers.
- **Technical communication:** translated the data into engineering presentations used in fundraising.

WHAT I TOOK FROM IT

- **Fixture design drives data quality.** In experimental work, repeatability is the whole game — small mechanical improvements to alignment had far more impact on confidence in the result than anything downstream in the analysis.
- **Constraints force decisions.** Limited budget and equipment meant choosing what actually mattered to measure well, rather than trying to control everything.

Technical Skills

Tools and methods I can speak to in detail and have applied on the work in this portfolio

CAD & Design

Autodesk Inventor · SolidWorks · Onshape · AutoCAD
GD&T (MMC/LMC) · worst-case tolerance stack-up · interference analysis · fixture design · engineering drawings · P&ID development

Programming & Analysis

Python (OpenCV, machine-vision inspection, pandas) · C++ · MATLAB · Excel
ANOVA / regression · I²C and PWM interfaces · Raspberry Pi · Arduino · hardware-software integration

Quality & Validation

Failure analysis · root-cause investigation · FMEA · dimensional inspection & metrology · experimental test design · nonconformance investigation · validation and test documentation

Fluid & Test Systems

Hydraulic/pneumatic architecture · pressure regulation & relief · closed-loop recirculation · pump and chiller sizing · manifold design · pressure-decay and flow-based leak test methods · test-stand architecture

Manufacturing & Prototyping

FDM 3D printing (PLA, PETG) · SLA/resin printing · rapid prototyping · mechanical assembly · fixture fabrication · design for manufacturing · design for assembly

Engineering Communication

Customer-facing failure-analysis reporting · design reviews · cross-functional requirements definition · vendor and supplier coordination · technical presentations to non-technical audiences

ADDITIONAL PROJECT EXPERIENCE

Assistive Sweeper Robot

Cornerstone II, London · Jan-Apr 2025

Co-designed an RC sweeper in SolidWorks, working drivetrain geometry against motor torque and ground clearance constraints. Iterated through a rapid prototyping sprint for operator usability and assembly.

Autonomous Maze Robot

Cornerstone I, Oakland · Sep-Dec 2024

Programmed sensor-based autonomous navigation logic. Diagnosed drivetrain binding and electrical contact failures through systematic root-cause analysis across a two-week test cycle.

Hydrogen Fuel Cell RC Vehicle

TKS Innovation Fellow · 2023-24

Designed a hydrogen fuel cell power system, managing power distribution and integration. Won a Google-sponsored pitch competition for the technical proposal.

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